

Cenk Baykal

ML & QUANT RESEARCHER

baykal@alum.mit.edu | [baykalc.github.io](https://github.com/baykalc) | linkedin.com/in/cenkbaykal | [Google Scholar](https://scholar.google.com/citations?user=...)

EDUCATION

Massachusetts Institute of Technology

PH.D. IN COMPUTER SCIENCE (GPA: 5.00/5.00)

Cambridge, MA

2017–2021

- Minor: Probability in High Dimension (Mathematics)
- Thesis: *Sampling-based Algorithms for Fast and Deployable AI*
- Advisor: Daniela Rus

Massachusetts Institute of Technology

S.M. IN COMPUTER SCIENCE (GPA: 4.91/5.00)

Cambridge, MA

2015–2017

- Thesis: *Algorithms for Persistent Autonomy and Surveillance*
- Advisor: Daniela Rus

University of North Carolina at Chapel Hill

B.S. COMPUTER SCIENCE WITH HIGHEST HONORS, B.A. MATHEMATICS (GPA: 3.91/4.00)

Chapel Hill, NC

2011–2015

- Graduated with Highest Distinction
- Thesis: *Design Optimization Algorithms for Concentric Tube Robots*
- Advisor: Ron Alterovitz

EXPERIENCE

Two Sigma

QUANTITATIVE RESEARCHER

New York City, NY

July 2024–Present

- Designed systematic alpha strategies that leverage alternative data sources and advanced machine learning techniques
- Built and maintained research tooling using agentic AI and LLMs to accelerate signal capture and forecast evaluation

Google Research

RESEARCH SCIENTIST

Cambridge, MA

January 2022–June 2024

- Developed conditional computation algorithms for transformers that enabled up to 30% speedups on large language models (LLMs)
- Designed data-efficient knowledge distillation strategies that led to improved transformer architectures with only 50% of teacher labeling cost
- Mentored scholars in Google CSRMP, supporting project design and research execution for students from historically marginalized groups
- Received the 2023 Google Research Tech Impact Award for leading “high-impact projects made sustainable by achieving tech excellence and great team dynamics”
- Received a Google Spot Bonus for “critical contributions to the efficiency of compact Gemini models”

Massachusetts Institute of Technology

POSTDOCTORAL RESEARCHER

Cambridge, MA

September 2021–January 2022

- Worked on algorithms for privacy-aware and efficient machine learning

J.P. Morgan AI Research

AI RESEARCH INTERN

New York City, NY

May 2021–September 2021

- Developed sampling-based algorithms with regret guarantees for large-scale graph neural network training

Massachusetts Institute of Technology

TEACHING ASSISTANT FOR ADVANCED ALGORITHMS (6.854J / 18.415J)

Cambridge, MA

Fall 2019

- Conducted office hours to help students on problem sets and concepts covered in lectures; designed and graded assignments
- TA rating according to the official MIT subject evaluation report: 7.0/7.0

Microsoft, SAS Institute, University of North Carolina at Chapel Hill

Redmond, WA; Cary, NC;
Chapel Hill, NC
2012–2015

EARLIER INDUSTRY & TEACHING EXPERIENCE

- Shipped SQL Server compression optimizations and analytics tooling deployed across academic and enterprise users
- Conducted research in robotics, path planning, and enabling technologies

HONORS & AWARDS

2024	Google Spot Bonus – Critical contributions to the efficiency of compact Gemini models	Google
2023	Google Research Tech Impact Award – Contributions to ML efficiency techniques	Google
2023	NeurIPS Spotlight Paper – Alternating Updates for Efficient Transformers	NeurIPS
2021	Winner – MIT The Engine's Interval Program (one of two winning teams)	MIT
2020–2021	Expert Reviewer – ICLR (2020, 2021); ICML (2021)	ICLR, ICML
2020	Top 10% Reviewer – NeurIPS	NeurIPS
2017	RSS Best Paper Award – Robotics: Science and Systems Conference	RSS
	Early Academic Honors – Carolina Research Scholar; CRA Outstanding Undergraduate	
2011–2015	Finalist; Phi Beta Kappa; Charles H. Dunham Scholarship; Dunlevie Honors Award; SURF; Honors Carolina; Dean's List	UNC Chapel Hill

SELECTED PUBLICATIONS

For the complete list of publications, visit [Google Scholar](#).

TRANSFORMERS & DISTILLATION

Alternating Updates for Efficient Transformers

CENK BAYKAL, DYLAN CUTLER, NISHANTH DIKKALA, NIKHIL GHOSH, RINA PANIGRAHY, XIN WANG

NeurIPS (Spotlight)
2023

SLaM: Student-Label Mixing for Distillation with Unlabeled Examples

VASILIS KONTONIS, FOTIS ILIOPOULOS, KHOA TRINH, CENK BAYKAL, GAURAV MENGHANI, ERIK VEE

NeurIPS
2023

Robust Active Distillation

CENK BAYKAL, KHOA TRINH, FOTIS ILIOPOULOS, GAURAV MENGHANI, ERIK VEE

ICLR
2023

A Theoretical View on Sparsely Activated Networks

CENK BAYKAL, NISHANTH DIKKALA, RINA PANIGRAHY, CYRUS RASHTCHIAN, XIN WANG

NeurIPS
2022

COMPRESSION & ROBOTICS

SiPPing Neural Networks: Sensitivity-informed Provable Pruning of Neural Networks

CENK BAYKAL*, LUCAS LIEBENWEIN*, IGOR GILITSCHENSKI, DAN FELDMAN, DANIELA RUS

SIAM SIMODS
2022

Lost in Pruning: The Effects of Pruning Neural Networks beyond Test Accuracy

LUCAS LIEBENWEIN, CENK BAYKAL, BRANDON CARTER, DAVID GIFFORD, DANIELA RUS

MLSys
2021

Provable Filter Pruning for Efficient Neural Networks

CENK BAYKAL*, LUCAS LIEBENWEIN*, HARRY LANG, DAN FELDMAN, DANIELA RUS

ICLR
2020

Data-Dependent Coresets for Compressing Neural Networks with Applications to Generalization Bounds

CENK BAYKAL*, LUCAS LIEBENWEIN*, IGOR GILITSCHENSKI, DAN FELDMAN, DANIELA RUS

ICLR
2019

Sampling-Based Approximation Algorithms for Reachability Analysis with Provable Guarantees

CENK BAYKAL*, LUCAS LIEBENWEIN*, IGOR GILITSCHENSKI, SERTAC KARAMAN, DANIELA RUS

Robotics: Science and Systems

2018

Asymptotically Optimal Design of Piecewise Cylindrical Robots using Motion Planning

CENK BAYKAL, RON ALTEROVITZ

Robotics: Science and Systems (Best Paper Award)
2017